Tax Policy Design in a Globalized Economy: A Comparative Analysis of Destination and Origin Principles.*

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Abstract

We study the optimal design of commodity taxation in a globalized economy with mobile firms and heterogeneous labor markets. We develop a two-country general-equilibrium model of trade that features monopolistic competition, endogenous firm relocation, and a dual labor market in which skilled workers earn flexible wages while unskilled workers face wage rigidity. Within this framework, we compare the welfare implications of two tax regimes: the destination principle (taxation at the point of consumption) and the origin principle (taxation at the point of production). Our analysis yields a sharp criterion: when the share of production income accruing to skilled labor exceeds 50%, the origin principle delivers higher equilibrium welfare. This finding provides a tractable decision rule based on observable labor market parameters and highlights how incorporating firm mobility and labor heterogeneity can overturn conventional wisdom, offering new insights for international tax policy.

Keywords: tax competition, origin principle, destination principle, monopolistic competition, labor heterogeneity.

JEL classification: F10, H20, H25.

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1 Introduction

Over recent decades, the deepening integration of national economies has intensified long-standing concerns about the strategic use of commodity taxes. Even in the absence of formal trade barriers, governments can adjust tax rates to attract mobile firms or cross-border shoppers, risking a "race to the bottom" that erodes public revenues and distorts the spatial allocation of economic activity. At the heart of this policy debate lies a fundamental institutional choice: should goods be taxed in the country of production (the origin principle) or in the country of consumption (the destination principle)?

Although countries have been operating destination-based value-added taxes (VAT), this consensus is increasingly strained by three interrelated developments. First, the removal of border controls within integrated economic areas, most notably the European Union's single market, has made cross-border shopping a first-order concern. When consumers can purchase freely across jurisdictions, destination-based taxation creates incentives for tax arbitrage that can undermine national fiscal autonomy (Asplund et al., 2007; Burstein et al., 2024). Second, the exponential growth of e-commerce has made traditional border tax adjustments increasingly difficult to enforce, as digital transactions bypass many of the physical checkpoints on which the destination-based collection relies (Aiura and Ogawa, 2023; Goolsbee, 2001). With global e-commerce sales projected to reach 22% of retail sales by 2026 (Statista, 2024a,b), these enforcement challenges will only intensify. Third, within the EU, deepening integration has been accompanied by persistent regional disparities in terms of the geographic distribution of firms, a concern explicitly recognized in the Amsterdam Treaty's cohesion objectives, raising the question of whether tax competition might amplify rather than mitigate spatial inequality (Behrens et al., 2009).

These pressures have revived a classic question in public finance: Which tax principle, destination or origin, is welfare-superior in an integrated economy? The academic literature provides no clear consensus. Early competitive models established a qualified presumption in favor of destination-based taxation, grounded in the Diamond-Mirrlees production efficiency theorem: destination-based taxes preserve equal marginal costs across locations, whereas origin-based taxes generally distort production patterns (Lockwood, 2001). However, this presumption has been challenged by imperfect competition models, which show that origin-based taxation can sometimes achieve first-best outcomes when firms have market power (Keen and Lahiri, 1998). More recent work incorporating firm mobility reveals that the two principles generate fundamentally different patterns of tax competition, with the origin principle triggering a more vigorous "race to the bottom" but also promoting a more equal spatial distribution of firms (Behrens et al., 2009).

This paper revisits the destination-origin debate by introducing a dimension that has been largely absent from existing analyses: labor market heterogeneity. We develop a two-country general equilibrium model featuring monopolistic competition, endogenous firm relocation, and crucially a dual labor market in which skilled workers earn flexible wages while unskilled workers face wage rigidity. This structure captures a salient feature of modern economies: the coexistence of flexible, high-skill labor markets and institutionally rigid low-skill labor markets, with the latter often absorbing demand shocks through quantity adjustments rather than price adjustments (Autor, 2014; Katz and Autor, 1999b).

Our central finding – our analysis been conducted in a symmetric case– is a sharp, empirically implementable criterion for the choice of tax principle. When the share of production income accruing to skilled labor exceeds 50%, the origin principle delivers higher equilibrium welfare. This threshold result emerges from the differential adjustment mechanisms activated by each tax regime. Under the origin principle, a tax increase raises domestic production costs, triggering firm relocation. This potential exodus depresses local demand for skilled labor, forcing the flexible skilled wage to adjust downward thus decreasing domestic production costs; the opposite happening in the foreign country. When production is skill-intensive, this wage compression significantly reduces firms' marginal costs, acting as an internal devaluation that stabilizes the domestic tax base and preserves product variety. Under the destination principle, by contrast, firm location is invariant to tax changes, taxes apply where goods are consumed, not where they are produced, so this adjustment channel is absent, and the full burden of taxation falls on output quantities.

The remainder of the paper is organized as follows. Section 3 presents the model. Section 4 derives the market equilibrium under the destination and the origin principles. Section 5 characterizes cooperative (joint-welfare-maximizing) tax outcomes. Section 6 analyzes non-cooperative tax competition and establishes the central threshold result. Section 7 calibrates the model using EU KLEMS data, showing that all EU member states lie above the 50% threshold under plausible skill definitions, implying that the origin principle would welfare-dominate the current destination-based VAT system. Section 8 concludes with implications for tax policy and suggestions for future research..

2 Related literature

Our analysis connects to several interconnected strands of research in public economics and international trade.

2.1 Tax Principles under Perfect Competition

The modern literature on destination versus origin taxation originates with analyzes of perfectly competitive economies. Under highly stylized conditions, uniform tax rates across countries, and full price flexibility, the two principles can be equivalent in their real effects (Grossman, 1980). However, this neutrality breaks down once governments set taxes non-cooperatively. Mintz and Tulkens (1986) provided the first systematic analysis of commodity tax competition, demonstrating that origin-based taxes yield an inefficient Nash equilibrium in which governments undercut each other to attract mobile shoppers. Lockwood (1993) extended this framework, showing that neither principle unambiguously dominates when countries can affect their terms of trade. A central insight from this literature is that destination-based taxation avoids cross-border fiscal externalities when countries are small, making it potentially more conducive to efficiency (Lockwood, 2001, Proposition 1). Our model builds on these foundations but introduces two features absent from competitive analyses: endogenous firm location and heterogeneous labor markets with differential wage flexibility.

2.2 Imperfect Competition and Tax Incidence

The recognition that imperfect competition fundamentally alters tax incidence prompted a re-evaluation of the destination-origin debate. Keen and Lahiri (1998) develop a two-country Cournot duopoly model with homogeneous goods and show that, contrary to competitive-case wisdom, origin-based taxation can achieve first-best outcomes while destination-based taxation cannot. The intuition is that origin taxes act directly on firms' output decisions, allowing governments to correct monopoly distortions through production subsidies, whereas destination taxes operate through consumption and cannot simultaneously address production inefficiency when firms have different costs. Our model shares with Keen and Lahiri (1998) the emphasis on imperfect competition but extends their analysis in three directions: (i) monopolistic competition with differentiated products and free entry, (ii) endogenous firm location across countries, and (iii) heterogeneous labor with segmented wage-setting mechanisms. Haufler and Pflüger (2004) analyze commodity taxation in a Dixit-Stiglitz monopolistic competition framework with iceberg trade costs and internationally mobile firms, the closest antecedent to our modeling approach. Their striking result is that non-cooperative taxation under the destination principle achieves Pareto efficiency, because positive and negative fiscal externalities exactly offset. Under the origin principle, on the contrary, additional externalities on the foreign tax base and price level lead to inefficiently high taxes. However, in our model with labor market segmentation, the efficiency properties

diverge: the destination principle no longer achieves Pareto efficiency, and the relative performance of the two regimes depends critically on the skilled labor share. This suggests that the Haufler and Pflüger (2004) neutrality result is not robust to the introduction of realistic labor market frictions.

2.3 Firm Mobility and Spatial Allocation

The "new economic geography" and related trade models have drawn attention to the role of firm location in shaping tax competition. Behrens et al. (2009) develop a model with mobile firms, trade frictions, and asymmetric country sizes to compare destination and origin principles. They find that origin-based taxation exacerbates tax competition and erodes revenues but promotes a more equal spatial distribution of industry, a direct implication of the "home market effect" being attenuated when larger countries charge higher taxes. Our model reproduces this spatial trade-off but adds an additional dimension: the incidence of tax-induced relocation on different worker types. Because skilled wages adjust flexibly while unskilled wages are rigid, the welfare consequences of firm relocation are distributed asymmetrically across skill groups. This distributional channel, absent in Behrens et al. (2009), generates our threshold condition linking optimal tax regimes to labor composition.

2.4 Labor Market Frictions and Tax Policy

A growing literature examines how labor market imperfections interact with international tax competition. Moriconi and Sato (2009) analyze commodity taxation in the presence of unemployment, showing that destination-based taxes can be used to manipulate the terms of trade and reduce unemployment, whereas origin-based taxes cannot. Antoniou et al. (2019) extend this analysis to large open economies, deriving conditions under which one principle dominates in the presence of labor market rigidities. Our approach differs in two key respects. First, rather than modeling unemployment, we assume a perfectly competitive numeraire sector that absorbs unskilled labor displaced from the differentiated sector. This captures "underemployment", labor relegated to lower-value activities, as the relevant margin of adjustment, consistent with evidence that employment-to-employment transitions, rather than unemployment, dominate labor market responses to demand shocks (Autor, 2014). Second, we explicitly model the wage flexibility differential between skill groups, which allows us to trace how tax-induced relocation affects the skill premium and the distribution of income.

3 The Model

We consider a version of the Dixit-Stiglitz-Krugman model of monopolistic competition (Dixit and Stiglitz, 1977; Krugman, 1979 and Krugman, 1980) where firms produce monopolistically goods that enter international trade using one unit of capital and a variable amount of labor. The economy is made up of two countries, home h and foreign f. We assume that the home and the foreign countries are populated by L and L^* individuals respectively. In addition, suppose that each country is populated by two types of individuals, skilled denoted by s and unskilled denoted by u. Let us further denote by ψ_i (resp. ψ_i^*) $i \in \{s, u\}$ the proportion of individuals of type i in the total population of the home country (resp. in the foreign country). Each individual owns one unit of labor and an amount of capital that differs from one type of individual to another while being the same for all individuals of the same type.

We assume that there are two goods: a numeraire produced exclusively by unskilled individuals, and the other, a differentiated good, produced by firms using both unskilled and skilled labor. We further suppose that firms produce goods using a Cobb-Douglas technology and these goods are sold domestically and abroad. A tax is applied to each unit of good so that when it becomes too high in a given country, some firms relocate abroad, in this paper, without costs. Finally, in each country, there is a local government that collects fiscal revenues and redistributes them to their citizens as a lump sum.

In what follows, we describe the economy and then discuss changes that occur when the level of the tax changes in one country according to the destination and origin principles.

3.1 Consumer demand

A given consumer of type i in the home country maximizes its utility under the tax principle k and defined on a set of differentiated goods $D_{i,k}$ and a numeraire $E_{i,k}$. We assume $D_{i,k}$ is produced by monopolistic firms located in the home and foreign countries and the numeraire $E_{i,k}$ is produced under perfect competition.

We further assume that the numeraire enters the utility function linearly. Therefore, the utility function of a type-i consumer is represented by the following equation :

$$U_{i,k} = \mu_i \ln D_{i,k} + E_{i,k}, \quad \mu_i > 0 \tag{1}$$

where

$$D_{i,k} = \left[\int_{\Omega} D^h_{i,k}(v)^{(\sigma-1)/\sigma} dv + \int_{\Omega^*} D^f_{i,k}(v)^{(\sigma-1)/\sigma} dv \right]^{\sigma/(\sigma-1)}, \ \sigma > 1. \tag{2}$$

$D^h_{i,k}$ is the demand for variety v produced in the home country h (resp. $D^f_{i,k}$ the demand for variety v produced in the foreign country f) of any consumer of type i located in the home country h. Ω (resp. Ω^*) is the number of varieties in the home country (resp. in the foreign country).

Let us assume that one unit of the numeraire is produced under constant return to scale and perfect competition by using one unit of unskilled labor. We further assume that this numeraire is not part of international trade.[1]. These two assumptions allow us to normalize the price of the numeraire and the wage of the unskilled workers.

For any differentiated good, an ad valorem tax is applied to the producer price regardless of destination or origin principles. Therefore, the budget constraint of a type-i consumer is given by:

$$\int_{\Omega} (1 + t_{h,k})\, p_{h,k}(v) D^h_{i,k}(v) dv + \int_{\Omega^*} (1 + t_{f,k})\, p_{f,k}(v) D^f_{i,k}(v) dv + E_{i,k} = \omega_{i,k} + R_k K_i + B_{i,k}. \tag{3}$$

K_i is the amount of capital supplied by the consumer of type i, R_k is the capital return under the tax principle k, and $B_{i,k}$ is a lump sum to any consumer of type i. In addition, $p_{h,k}$ (resp. $p_{f,k}$) represents the producer price for goods produced in the home country (resp. in the foreign country).[2]

Introducing the price index P_k,

$$P_k = \left(\int_{\Omega} \left[(1 + t_{h,k})\, p_{h,k} \right]^{1-\sigma} + \int_{\Omega^*} \left[(1 + t_{f,k})\, p_{f,k} \right]^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \tag{4}$$

the above budget constraint in the home country h can be re-written as

$$P_k D_{i,k} + E_{i,k} = \omega_{i,k} + R_k K_i + B_{i,k}, \qquad k \in \{d, o\}. \tag{5}$$

We can now express the demand for domestic and foreign goods as a function of tax levels

[1]The numeraire is often conceptualized as a non-tradable service or good (e.g., local haircuts, real estate services, construction) whose international exchange is either impossible or prohibitively costly. This assumption is standard in international trade and tax competition models, particularly those that aim to simplify general equilibrium dynamics while isolating the effects of tax policy on traded goods; see Grossman (1980) or Helpman and Krugman (1985). See Melitz (2003) or Costinot and Rodríguez-Clare (2014) for cases where the numeraire is tradable.

[2]In the foreign country f, will be denoting respectively $p^*_{h,k}$ and $p^*_{f,k}$ the producer prices for goods produced in the home country h and in the foreign country f respectively.

and the price index, for a consumer located in the domestic country h, as follows:

$$D_{i,k}^{h} = \mu_i \left[(1+t_{h,k})\, p_{h,k}\right]^{-\sigma} P_k^{\sigma-1}; \quad D_{i,k}^{f} = \mu_i \left[(1+t_{f,k})\, p_{f,k}\right]^{-\sigma} P_k^{\sigma-1} \tag{6}$$

Using the same reasoning, the domestic and foreign demand of any consumer of type i located in the foreign country are given by:

$$D_{i,k}^{h,*} = \mu_i \left[\left(1+t_{h,k}^{*}\right) p_{h,k}^{*}\right]^{-\sigma} (P_k^{*})^{\sigma-1}; \quad D_{i,k}^{f,*} = \mu_i \left[\left(1+t_{f,k}^{*}\right) p_{f,k}^{*}\right]^{-\sigma} (P_k^{*})^{\sigma-1}.^{3} \tag{7}$$

$D_{i,k}^{h,*}$ (resp. $D_{i,k}^{f,*}$) is the demand for the variety v produced in the domestic country (resp. produced in the foreign country) by an individual of type i located in the foreign country.

The indirect utility function in the domestic country is then given by:

$$V_{i,k}\left(t_{h,k}, t_{f,k}\right) = \mu_i \ln\left(\mu_i P_k^{-1}\right) + \left(\omega_{i,k} + R_k K_i + B_{i,k} - \mu_i\right), \; k \in \{d, o\} \tag{8}$$

3.2 The production

A firm located in the home country h uses skilled and unskilled labor to produce goods that will be sold on the domestic and foreign markets. We assume a Cobb-Douglas production function represented as follows:

$$\sum_i \psi_i D_{i,k}^{h}(v) L + \sum_i \psi_i^{*} D_{i,k}^{h,*}(v) L^{*} = F_k\left(\ell_{s,k}, \ell_{u,k}\right) \equiv A \ell_{s,k}^{\alpha} \ell_{u,k}^{1-\alpha}, \tag{9}$$

where $\ell_{i,k}$ is the amount of labor of type i that is used by firms in their production process under the tax principle k.
The corresponding technological constraint for any firm located in the foreign country is given by :

$$\sum_i \psi_i D_{i,k}^{f}(v) L + \sum_i \psi_i^{*} D_{i,k}^{f,*}(v) L^{*} = F_k\left(\ell_{s,k}^{*}, \ell_{u,k}^{*}\right) \equiv A \left(\ell_{s,k}^{*}\right)^{\alpha} \left(\ell_{u,k}^{*}\right)^{1-\alpha} \tag{10}$$

In addition, suppose that prior to operation, each firm uses one unit of capital as fixed costs (those fixed costs encompass costs of building new factory, new machines, R&D and other

[3] Where the price index in the foreign country is given by

$$P_k^{*} = \left(\int_{\Omega} \left[\left(1+t_{h,k}^{*}\right) p_{h,k}^{*}\right]^{1-\sigma} + \int_{\Omega^{*}} \left[\left(1+t_{f,k}^{*}\right) p_{f,k}^{*}\right]^{1-\sigma}\right)^{\frac{1}{1-\sigma}}, \; k \in \{d, o\}$$

needs); this unit of capital is paid back in a return denoted R_k. Further, assume that capital is freely mobile between countries without any additional costs; this assumption equalizes the price of capital in the two countries. In addition, we assume that for each unit of skilled labor used by the firms, the firms pay a cost equal to ω_k which represents the skilled wage. Therefore, the firm that produces variety w in the country h maximizes its profits given by:

$$\pi_{h,k}(v) = p_{h,k}(v) \sum_i \psi_i D^h_{i,k}(v) L + p^*_{h,k}(v) \sum_i \psi^*_i D^{h,*}_{i,k} L^* - \omega_k \ell_{s,k}(v) - \ell_{u,k}(v) - R_k, \quad (11)$$

Under monopolistic competition, each firm sets the domestic and foreign prices, $p_{h,k}(v)$ and $p^*_{h,k}(v)$, which maximize its profit in each market. Therefore from the first-order conditions we have $p_{h,k}(v) = p^*_{h,k}(v)$; and using the technological constraint (9), we can re-write firms' profit function as follow:

$$\pi_{h,k}(v) = \frac{p_{h,k}(w) F_k(w)}{\sigma} - R_k \quad (12)$$

The first term on the right-hand side of equation (12) represents firm's operating surplus i.e. the proportion of total revenue that is left after covering for variable costs (here labor). There will be an increasing number of firms entering the market until this terms exactly offset the fixed costs R_k.

Assuming no firm earns profit, therefore, the remuneration of skilled and unskilled labor used in the production process is given by the following:

$$\omega_k \ell_{s,k}(v) = \alpha(\sigma - 1) R_k; \quad \ell_{u,k}(v) = (1 - \alpha)(\sigma - 1) R_k.^{45} \quad (13)$$

Equation (13) show how firms distribute their markups (returns above cost) between skilled and unskilled labor. Skilled workers receive a flexible wage that adjusts with profitability, while unskilled labor is hired in quantities that vary with firm income due to fixed wages. To be more precise, as markup (i.e., market power or demand intensity) or return on capital increases, firms pay more to skilled labor, but this payment is constrained by the elasticity of skilled worker to the production α; in the other hand, as firms become more profitable (higher R_k), they hire more unskilled labor in fixed proportion.

[4]The corresponding equations in the foreign country are given by:

$$\omega^*_k \ell^*_{s,k}(v) = \alpha(\sigma - 1) R_k; \quad \ell^*_{u,k}(v) = (1 - \alpha)(\sigma - 1) R_k.$$

[5]$(\sigma - 1) R_k$ represents markup revenue above marginal cost, i.e., the total funds available for variable input payments (here in this paper, labor) after the firm covers its fixed cost R_k.

3.3 The Government

Now we will give a proper expression to the income that comes from taxes and is distributed to consumers as a lump sum. As a reminder, in our paper, the sole purpose of governments is to maximize the welfare of consumers in their respective countries. Governments therefore set the level of tax paid per unit of good consumed, and this fiscal revenue is collected and passed on to all citizens in the form of lump sum transfers. In the destination principle, taxes are collected on goods sold on the domestic market (locally and imported goods), while in the origin principle, taxes are collected on goods locally produced (whether sold domestically or exported).

Therefore, the fiscal revenue in the home and foreign countries in the destination principle are given by:

$$\begin{cases} t_d T_d = t_d L \sum_i \psi_i \left(N_d p_{h,d} D^h_{i,d} + N^*_d p_{f,d} D^f_{i,d} \right) & = L \sum_i \psi_i B_{i,d} \\ t^*_d T^*_d = t^*_d L^* \sum_i \psi^*_i \left(N_d p^*_{h,d} D^{h,*}_{i,d} + N^*_d p^*_{f,d} D^{f,*}_{i,d} \right) & = L^* \sum_i \psi^*_i B^*_{i,d} \end{cases} \tag{14}$$

While in the origin principle, fiscal revenues in the home and foreign countries are:

$$\begin{cases} t_o T_o = t_o N_o \left[L \sum_i \psi_i p_{h,o} D^h_{i,o} + L^* \sum_i \psi^*_i p^*_{h,o} D^{h,*}_{i,o} \right] & = & L \sum_i \psi_i B_{i,o} \\ t^*_o T^*_o = t^*_o N^*_o \left[L \sum_i \psi_i p_{f,o} D^f_{i,o} + L^* \sum_i \psi^*_i p^*_{f,o} D^{f,*}_{i,o} \right] & = & L^* \sum_i \psi^*_i B^*_{i,o} \end{cases} \tag{15}$$

where $T_k, k \in \{d, o\}$ (resp. $T^*_k, k \in \{d, o\}$) is the fiscal base under the tax principle k in the domestic country (resp. in the foreign country).

4 Equilibrium

In this Section we give the remaining equations that characterize the equilibrium in the model and the hypothesis associated to each of them.

- Labor market equilibria : we assume in our model that there is no international mobility of skilled labor; therefore, the supply of skilled labor in a given country is equal to the amount of skilled labor effectively used by firms located in that country. This

characterization is given by:

$$\psi_s L = N_k \ell_{s,k}; \qquad \psi_s^* L^* = N_k^* \ell_{s,k}^* \tag{16}$$

The previous equations state that the number of companies existing in the two countries depends inversely on the quantity of skilled labor used: the greater the quantity of skilled labor used, the smaller the number of companies existing.

- Capital market equilibrium: each firm used one unit of capital as fixed costs and capital is freely mobile between countries. Therefore the amount of capital in the economy determines the overall existing number of firms; this is characterized by:

$$N_k + N_k^* = \bar{K} \tag{17}$$

- Free-Entry Conditions: under monopolistic competition, firms enter markets until profits fall to zero; the entry of each new firm diminishes the profits of existing ones until revenue exactly balances fixed costs. This is therefore represented as:

$$p_{f,k} A \left(\ell_{s,k}^*\right)^{\alpha} \left(\ell_{u,k}^*\right)^{1-\alpha} = p_{h,k} A \left(\ell_{s,k}\right)^{\alpha} \left(\ell_{u,k}\right)^{1-\alpha} = \sigma R_k \tag{18}$$

From now on, we will refer to ω_k $k \in \{d, o\}$ as the wage rate of the skilled worker. Indeed, since the unskilled wage rate is constant and equal to one, thus not subject to tax variation, this will ease our notation. In addition, we choose to fix to a constant the redistribution from tax collection B_u and B_u^* that goes to the unskilled workers both in the home and in foreign countries.

Therefore the two price indexes can be re-expressed as:

$$\begin{cases} P_k &= \left[N_k \left[(1+t_{h,k})\, p_{h,k}\right]^{1-\sigma} + N_k^* \left[(1+t_{f,k})\, p_{f,k}\right]^{1-\sigma}\right]^{\frac{1}{1-\sigma}}, \ k \in \{d, o\} \\ P_k^* &= \left[N_k \left[\left(1+t_{h,k}^*\right) p_{h,k}\right]^{1-\sigma} + N_k^* \left[\left(1+t_{f,k}^*\right) p_{f,k}\right]^{1-\sigma}\right]^{\frac{1}{1-\sigma}}, \ k \in \{d, o\} \end{cases} \tag{19}$$

where N_k (resp. N_k^*) is the number of companies located in the domestic country (resp. in the foreign country).[6]

[6] Using (13) and its foreign counter-part, and the free-entry conditions (18), we can derive producer prices

4.1 Destination principle

Under the destination principle, commodity taxes are levied in the country where goods and services are consumed, regardless of where they were produced. This approach is consistent with the destination-based VAT system used by most countries. It preserves competitive neutrality by ensuring that imported and domestically produced goods are taxed at the same rate. So we have, in the domestic country h, $t_{h,d} = t_{f,d}$, and in the foreign country f, $t^*_{h,d} = t^*_{f,d}$. In the remainder of this paper, when we are dealing with the destination principle, we will adopt the following notation $t_{h,d} = t_{f,d} = t$ and $t^*_{h,d} = t^*_{f,d} = t^*$.

Using the labor demand market (16) and the capital market equilibrium (17), we obtain the domestic and foreign number of firms as following:

$$N_d = \frac{\bar{K}}{1 + \frac{\psi_s^* L^*}{\psi_s L}}; \qquad N_d^* = \frac{\bar{K}}{1 + \frac{\psi_s L}{\psi_s^* L^*}} \tag{20}$$

In the destination principle, the number of firms is independent of the level of tax rate; that is, they are not mobile whatever the level of tax. They are solely dependent on the ratio of the number of skilled labor available in a given country to the total number of skilled labor. *Ceteris Paribus* , the country with the largest population of qualified workers, attracts the largest number of firms.

Using the the technological constraint and the free-entry condition, one gets the expression of the capital return as:

$$R_d = \frac{1}{\sigma \bar{K}} \left[\frac{L \sum_i \psi_i \mu_i}{1+t} + \frac{L^* \sum_i \psi_i^* \mu_i}{1+t^*} \right] \tag{21}$$

Using (13), (16) and (20), the wage of any skilled worker located abroad or in the domestic

at home and abroad as follows:

$$\begin{aligned} p_{h,k} = p^*_{h,k} &= \frac{\sigma}{\sigma - 1} \frac{\omega_k^\alpha}{A(\alpha)^\alpha (1-\alpha)^{1-\alpha}} \\ p_{f,k} = p^*_{f,k} &= \frac{\sigma}{\sigma - 1} \frac{(\omega_k^*)^\alpha}{A(\alpha)^\alpha (1-\alpha)^{1-\alpha}} \end{aligned}$$

These expressions show that the pre-tax producer prices are proportional to marginal cost (which depends on wages) with the same markup $\frac{\sigma}{\sigma-1}$ in both tax regimes. Notably, no explicit tax term appears in the producer price formula, indicating that firms choose their pre-tax price according to the usual markup rule despite the presence of ad valorem taxes. In other words, a consumption tax or production tax does not alter the markup pricing condition directly; instead, taxes influence producer prices indirectly through general-equilibrium effects on wages and demand.

country is given by:

$$\omega_d^* = \omega_d = \frac{\alpha}{L\psi_s + L^*\psi_s^*}\frac{\sigma-1}{\sigma}\left[\frac{L\sum_i \psi_i\mu_i}{1+t} + \frac{L^*\sum_i \psi_i^*\mu_i}{1+t^*}\right] \tag{22}$$

The wage rate as well as the capital return are decreasing functions of the tax rate. The intuition is that the increase in tax increases the consumer price, thus reducing the demand. This reduction of the demand of commodity goods has two effects: first, it decreases the demand of labor (both skilled and unskilled); but since the supply of skilled labor remains unchanged, the adjustment comes by decreasing the wage rate of the skilled labor. Second, it decreases the demand for capital, and since the supply is unchanged, then the capital return decreases to adjust the gap.

Finally, the fiscal base in the domestic country h and its foreign counterpart in the foreign country f are given by

$$T_d = \frac{L\sum_i \psi_i\mu_i}{1+t}; \qquad T_d^* = \frac{L^*\sum_i \psi_i^*\mu_i}{1+t^*} \tag{23}$$

In the destination principle, the tax revenue in a given country solely depends on the tax rate applied in this country : the greater the tax level (small subsidy), the lower the fiscal base. This is due to the fact that in the destination, tax directly target the consumer by increasing the consumer price, thus reducing the demand, further increasing the distortion due by the monopolistic competition.

4.2 Origin principle

The origin principle applies taxes in the country where goods and services are produced. Exports are taxed, while imports are exempt. This approach ties tax policy directly to the production location, and thus influences firms' location decisions. Therefore in each country, we have $t_{h,k} = t^*_{h,k}$ in the domestic country h and $t_{f,k} = t^*_{f,k}$ in the foreign country f. Thus, according to the origin principle, the tax is discriminatory and can be actively used as a tool to benefit domestic producers. In the remainder of this paper, when we are dealing with the origin principle, we will adopt the following notation : $t_{h,k} = t^*_{h,k} = t$ and $t_{f,k} = t^*_{f,k} = t^*$.

From the notations in this section and equations (4) and (3), the consumer price indexes in the domestic and foreign countries are exactly the same

Using the two labor demand equations in (16) and the capital market equation (17), we have

$$N_o = \frac{\bar{K}}{1+B}; \qquad N_o^* = \frac{\bar{K}B}{1+B} \tag{24}$$

where, $B = \frac{\psi_s^* L^*}{\psi_s L} \left(\frac{1+t^*}{1+t}\right)^{-\frac{\sigma}{\alpha(\sigma-1)}}$.

The term B captures the relative attractiveness of the foreign country for the location of the firm, as seen from the perspective of the home country such that $\frac{1}{1+B}$ and $\frac{B}{1+B}$ represent the shares of firms located in the home and foreign country, respectively; this structure also governs how firm distribution affects factor demand and wages.[7] To be more precise, B is the relative supply of skilled labor in the foreign country, adjusted for by the tax disadvantage of being in the foreign country. It is made up of two terms:

- $\frac{\psi_s^* L^*}{\psi_s L}$, is the relative size of the skilled labor force in the foreign country compared to that in the home country: the country with the most abundant population of skilled workers attracts more firms.
- $\left(\frac{1+t^*}{1+t}\right)^{-\frac{\sigma}{\alpha(\sigma-1)}}$, captures the relative tax-induced distortion on firm profitability; it reflects how a firm's optimal location decision is distorted by tax differentials between countries: for instance, if the domestic (resp. the foreign) country raises its tax level t (resp. t^*), this raises (resp. decreases) the relative attractiveness term B.

Using the free-entry equation and the expression of the price index, the capital price in the origin principle is given by :

$$R_o = \left(1 + \frac{L^* \sum_i \psi_i^* \mu_i}{L \sum_i \psi_i \mu_i}\right) \frac{L \sum_i \psi_i \mu_i}{\bar{K}\sigma\,(1+t)} \frac{1+B}{1+B\left(\frac{1+t^*}{1+t}\right)}, \tag{25}$$

where:

- The total demand for differentiated goods in the whole economy represented by $\left(1 + \frac{L^* \sum_i \psi_i^* \mu_i}{L \sum_i \psi_i \mu_i}\right) L \sum_i \psi_i \mu_i$. Intuitively, the more consumers value variety (higher μ_i) and the larger the population $(L + L^*)$, the more firms can sell, and the higher the return to capital.
- The capital cost per variety denoted here by $\bar{K}\sigma\,(1+t)$; this term captures the difficulty in generating a return to capital per variety in a given tax and competition

[7] This mechanism mirrors the FDI-driven relocation channel emphasized by McCracken (2015), who demonstrates that differences in commodity tax bases distort horizontal FDI flows by altering the profitability of operating across jurisdictions.

environment.[8]

Using (13) and its foreign counterpart, the expressions of the wage rate of skilled workers in the origin principle are given by:

$$\begin{cases} \omega_o &= \frac{\alpha(\sigma-1)}{L\psi_s\sigma}\left(1+\frac{L^*\sum_i\psi_i^*\mu_i}{L\sum_i\psi_i\mu_i}\right)\frac{L\sum_i\psi_i\mu_i}{1+t}\left[\frac{1}{1+B\left(\frac{1+t^*}{1+t}\right)}\right] \\ \omega_o^* &= \frac{\alpha(\sigma-1)}{L\psi_s\sigma}\left(1+\frac{L^*\sum_i\psi_i^*\mu_i}{L\sum_i\psi_i\mu_i}\right)\frac{L\sum_i\psi_i\mu_i}{1+t}\left[\frac{\left(\frac{1+t^*}{1+t}\right)^{-\frac{\sigma}{\alpha(\sigma-1)}}}{1+B\left(\frac{1+t^*}{1+t}\right)}\right] \end{cases} \tag{26}$$

The asymmetry in skilled wages across countries stems entirely from the term $\left(\frac{1+t^*}{1+t}\right)^{-\frac{\sigma}{\alpha(\sigma-1)}}$, which appears only in the foreign wage expression. This expression reflects how differences in production taxes translate into variations in the effective surplus available to foreign firms for skilled labor, with the magnitude of this adjustment shaped by both the elasticity of substitution, σ, and the skilled labor share in the production function, α.

Recalling that the two price indexes are equal in the origin principle, we have the following expressions of the fiscal bases in both countries:

$$\begin{cases} T_o &= \frac{L\sum_i\psi_i\mu_i+L^*\sum_i\psi_i^*\mu_i}{(1+t)\left[1+B\left(\frac{1+t^*}{1+t}\right)\right]} \\ T_o^* &= \frac{L\sum_i\psi_i\mu_i+L^*\sum_i\psi_i^*\mu_i}{1+t^*}\frac{B\left(\frac{1+t^*}{1+t}\right)}{1+B\left(\frac{1+t^*}{1+t}\right)} \end{cases} \tag{27}$$

As the attractiveness of the foreign country increases, the fiscal base decreases in the domestic country but increases in the foreign country as some domestic firms relocate abroad. This is clearly revealed in equation (24).

5 Cooperative optimum

In this section, we derive the optimal tax formulae that characterize the cooperative case; it will be used as a benchmark case. The analyzes will be performed in a symmetric case; therefore, the optimal tax formulae can be derived by maximizing the joint utilities of the two countries; that is to maximize the sum of the indirect utility in the domestic country h

[8]Since in our model the available amount of capital is $\bar{K}$, it also represents the total number of varieties in this economy since the production of one variety requires one unit of capital. But since each variety is less profitable both under high substitution (high σ) and high tax level, the term $\bar{K}\sigma(1+t)$ can reasonably be understood as the effective capital cost of maintaining one variety in the market.

expressed in (8) and its foreign counterpart. The first-order condition is given by :

$$L\sum_i \frac{\partial V_{i,k}}{\partial t} + L^*\sum_i \frac{\partial V^*_{i,k}}{\partial t} = 0. \tag{28}$$

The effects that must be taken into account when changing the level of the commodity tax in the domestic country h are given by:

$$L\sum_i \frac{\partial V_{i,k}}{\partial t} := -\left(L\sum_i \psi_i\mu_i\right)\frac{\partial P_k/\partial t}{P_k} + L\psi_s\frac{\partial \omega_k}{\partial t} + \frac{\partial R_k}{\partial t}\left(L\sum_i \psi_i K_i\right) + L\psi_s\frac{\partial B_{s,k}}{\partial t} \tag{29}$$

For the government of the domestic country h, before taking the decision to change the existing tax level, it will have to consider the effects that this variation will have on the four terms of equation (29). *Ceteris paribus*, the first term captures the variation in welfare following a variation in private consumption. The second and third terms taken together capture the variation in welfare due to a variation in income; the second (resp. third) term being the variation in labor income (resp. capital income). Finally, the last term captures the effect of variation in public consumption (or, more specifically, variation in consumption due to government transfers) on welfare. It is therefore important to determine the effects of a tax variation on the elements described in the equation and its foreign counterpart.

5.1 Destination principle

In this subsection, we give a formula for the optimal tax in the cooperative when the destination principle is used. For this purpose, we analyze how any tax variation affects the different variables and aggregate in the two countries.

In the destination principle, taxes are applied by the government of the country where the final good is consumed; therefore, we have $t_{h,d} = t_{f,d}$ in the domestic country h and $t^*_{h,d} = t^*_{f,d}$ in the foreign country f.

From (20) and (16)we have :

$$\frac{\partial N_d}{\partial t} = \frac{\partial N^*_d}{\partial t} = 0; \qquad \frac{\partial \ell_{s,d}}{\partial t} = \frac{\partial \ell^*_{s,d}}{\partial t} = 0 \tag{30}$$

Equation (30) shows that, in line with Behrens et al. (2009), destination-based taxation leaves the distribution of firms and the demand for skilled labor entirely unaffected by unilateral tax changes. Destination taxation therefore neutralizes strategic relocation responses and

eliminates agglomeration-driven tax competition, reflecting its key advantage over the origin principle.

Using the two price indexes in (19) the effect of any tax variation in the domestic country on the two price indexes are given by:

$$\frac{\partial P_d}{\partial t} = P_d \left[\frac{1}{1+t} + \frac{\alpha}{\omega_d} \frac{\partial \omega_d}{\partial t} \right]; \quad \frac{\partial P_d^*}{\partial t} = P_d^* \left(\frac{\alpha}{\omega_d} \frac{\partial \omega_d}{\partial t} \right) \tag{31}$$

Equation (31) gives the mechanism that influences the prices of home (first equation) and foreign (second equation). Any variation in the tax influences domestic prices in two ways: the first term reflects the direct effect of a variation in the tax level, and the second term captures the influence on prices in the two countries of the producer price spillovers. The effects of any variation in the tax on foreign prices are totally captured by the producer price spillovers, but on the domestic prices, a comparison needs to be done between the direct and indirect effects to see how prices evolve.

Using (13), (21), (22), (23), the effect of any tax variation in the domestic country on the skilled worker wages, capital return, tax revenue and the amount of unskilled labor is given by:

$$\frac{\partial \omega_d^*}{\partial t} = \frac{\partial \omega_d}{\partial t} = -\frac{\alpha(\sigma-1)}{\sigma} \frac{\sum_i \psi_i \mu_i}{2\psi_s (1+t)^2} < 0, \qquad \frac{\partial R_d}{\partial t} = -\frac{1}{\sigma \bar{K}} \frac{L \sum_i \psi_i \mu_i}{(1+t)^2} < 0 \tag{32}$$

$$L\psi_s \frac{\partial B_{s,d}}{\partial t} = \frac{L \sum_i \psi_i \mu_i}{(1+t)^2} > 0, \qquad L^* \psi_s^* \frac{\partial B_{s,d}^*}{\partial t} = 0 \tag{33}$$

The increase in the tax level in the home country targets in the same way domestic and foreign firms that sell in the home country. The increase in the commodity tax in the home country makes the domestic market less profitable for both domestic and foreign firms: capital return decreases, and firms reduce their demand for labor, leading to the fall of the wage of skilled labor as depicted in (32). Since the amount of skilled labor used in the production remains unchanged, firms reduction of labor simply translates into the reduction of the amount of unskilled labor used in the equilibrium; the quantity of unskilled labor removed from the production of the differentiated good is automatically returned to the production of the numeraire, thus satisfying the full employment of unskilled labor. Fiscal revenue increases in the domestic country but is unchanged abroad since there is no tax variation abroad and there is no firm's relocation.

Replacing (31), (32) and (33) into (28), we obtain the following.

$$-\frac{L\sum_i \psi_i \mu_i}{(1+t)^2}\left[(1+t)(1-\alpha)+\frac{\alpha(\sigma-1)}{\sigma}+\frac{1}{\sigma}-1\right]=0$$

$$\hat{t}=-\frac{1}{\sigma}$$

5.2 Origin principle

In the origin principle, the changes that occur in the home and foreign countries due to a variation in the level of tax in the home country are represented as follows in the symmetric equilibrium.

Using the expression of dual price (4), and the free entry condition (18), in symmetric equilibrium, we can derive the changes in aggregate consumer price due to the change of tax in the domestic country h as follows:

$$\frac{\partial P_o^*}{\partial t}=\frac{\partial P_o}{\partial t}=P_o\left[\frac{1}{1+t}\left(1+\frac{1}{2(\sigma-1)}\right)+\frac{\alpha}{\omega_o}\frac{\partial \omega_o}{\partial t}\right] \tag{34}$$

According to the principle of origin, a tax increase in the domestic country h increases the general price level on the domestic and foreign markets through direct and indirect effects. The direct effect is represented by the first term in brackets on the right-hand side of (34); It is made up of the direct effect of the tax on consumer prices and the effect of the change in the ratio of the number of firms in the foreign country to the number in the domestic country, is latter effect is not present in (31) since in the destination principle there is no relocation of firms abroad. In our model, the weight of domestic consumption is greater than 1, thus multiplying the direct effect of the tax. The second term in brackets reflects the indirect effect of the tax on the general price level; more precisely, this term shows that the price level is positively correlated with the wages of skilled workers; indeed, any tax that increases wages causes demand and, therefore, prices to rise.

$$\frac{\partial N_o}{\partial t}=-\frac{\sigma}{\alpha(\sigma-1)}\frac{\bar{K}}{4(1+t)}<0 \tag{35}$$

$$\frac{\partial \ell_{s,o}}{\partial t}=\frac{\sigma}{\alpha(\sigma-1)}\frac{\psi_s L}{\bar{K}}\frac{1}{1+t}>0; \qquad \frac{\partial \ell_{s,o}^*}{\partial t}=-\frac{\sigma}{\alpha(\sigma-1)}\frac{\psi_s L}{\bar{K}}\frac{1}{(1+t)}<0 \tag{36}$$

$$\frac{\partial \omega_o}{\partial t} = -\frac{\sum_i \psi_i \mu_i}{2\psi_s (1+t)^2}\left(1 + \frac{\alpha(\sigma-1)}{\sigma}\right) < 0; \quad \frac{\partial \omega_o^*}{\partial t} = \frac{\sum_i \psi_i \mu_i}{2\psi_s (1+t)^2}\left(1 - \frac{\alpha(\sigma-1)}{\sigma}\right) > 0 \tag{37}$$

Any tax increase in the home country h under the origin principle reduces the profitability of firms producing in the country h on this domestic market h and on foreign market f, causing some of them to relocate, as shown by (35); this result is similar to the one in Behrens et al. (2009) that states that the origin principle favors firms' spatial distribution. The remaining firms are reducing their demand for labor, especially skilled labor (36). Since, in our model, the supply of skilled labor is fixed and fully utilized in the production of differentiated goods, the fall in demand for skilled labor translates into a decrease in the wages of skilled workers as in (37). In contrast, abroad, the influx of companies from the domestic country h leads to an increase in labor demand and consequently to a rise in the wages of skilled workers, as shown by equations (36) and (37). Therefore the amount of skilled labor effectively used in the domestic h country increases while it decreases in the foreign country f as given in equation (36).

Another useful comment is that wage variation of skilled worker in the domestic and foreign countries are not exactly symmetrical even though the demand for skill labor and the movement of firms are symmetrical in those countries. In the domestic country h, the wages of skilled workers are influenced by the fall in corporate profitability resulting from higher prices –second term into parenthesis–, and also because of the relocation of some firms to the foreign country f –first term into parenthesis. In the foreign country f, two opposing phenomena influence the wage level of skilled workers: the rise in consumer prices due to higher taxes in the domestic country h, –first term into parenthesis–but also an increase in demand for skilled labor due to an increase in the number of firms –second term into parenthesis.

Equation (37) also clarifies how a domestic tax increase under the origin principle reshapes the wage differential between skilled and unskilled workers, a mechanism central to both our research question and the literature gap identified in the introduction. Because unskilled wages are fixed in the numeraire sector while skilled wages adjust flexibly, the tax-induced fall in domestic skilled wages and rise in foreign skilled wages (equation 37) widens the skilled-unskilled wage gap abroad and compresses it at home, whereas under the destination principle skilled wages remain unchanged (equation (30)). This asymmetric response, absent in models with homogeneous labor, shows how tax policy interacts with labor market segmentation to generate distributional consequences across worker groups, thereby filling the gap in earlier analyses that could not trace wage inequality or incidence across skill types. Finally, comparing (37) to (32), it is easy to see that any variation on the level of tax in

the domestic country has a higher negative effect on the level of wage of the skill worker in the origin principle than in the destination principle, this is simply due to the fact that the negative effect on the wage of skill worker due to firms' relocation abroad in the origin principle is nonexistent in the destination principle.

$$\frac{\partial R_o}{\partial t} = -\frac{L\sum_i \psi_i \mu_i}{\sigma \bar{K}\left(1+t\right)^2} < 0 \tag{38}$$

It is not surprising to see that the price of the capital falls since it is a good that is freely mobile across countries. Since firm's profitability falls for those located in the home country, this induces a decrease of the number of firms thus a reduction of the capital return. The mobility of capital thus ensures the equality of capital price in both countries. It's worth mentioning that the reduction in return on capital is exactly the same as in the destination principle.

In addition, the number of unskilled labor is decreasing in both countries. In the domestic country, this decline is linked to a fall in demand for labor due to a decrease in profitability. This fall in demand for labor has repercussions on both the quantity of skilled and unskilled labor used. However, in other countries, the fall in the unskilled labor used is due to an increase in the skilled labor used.

$$\frac{\partial T_o}{\partial t} = -\left(1+\frac{\sigma}{\alpha\left(\sigma-1\right)}\right)\frac{L\sum_i \psi_i \mu_i}{2\left(1+t\right)^2}; \qquad \frac{\partial T_o^*}{\partial t} = \left(\frac{\sigma}{\alpha\left(\sigma-1\right)}-1\right)\frac{L\sum_i \psi_i \mu_i}{2\left(1+t\right)^2} \tag{39}$$

$$L\psi_s \frac{\partial B_s^o}{\partial t} = \frac{L\sum_i \psi_i \mu_i}{\left(1+t\right)^2}\left[1-\frac{t}{2}\left(\frac{\sigma}{\alpha\left(\sigma-1\right)}-1\right)\right]; \quad L\psi_s \frac{\partial B_s^{o,*}}{\partial t} = \frac{L\sum_i \psi_i \mu_i}{\left(1+t\right)^2}\left[\frac{t}{2}\left(\frac{\sigma}{\alpha\left(\sigma-1\right)}-1\right)\right] \tag{40}$$

From (39), following an increase in the level of tax in the domestic country, the tax base decreases in that country but increases abroad. This is the simple result of the relocation to the foreign country f of some firms that had previously been located in the domestic country h due to the decrease in profitability. In the domestic country h, the tax revenue varies due to two effects: From the first equation in (40), the first term in brackets captures the increase in tax revenue due to the increase in the level of tax, while the second term captures the loss of tax revenue due to the fall of the fiscal base. This loss of revenue due to the reduction of the fiscal base in the domestic country is exactly what is gained in the foreign country, so these two effects cancel each other out.

Replacing (34), (37), (38) and (40) into (28), we have :

$$-\frac{L\sum_i \psi_i \mu_i}{2(1+t)^2}\left[2(1+t)(1-\alpha)+\left(1+\frac{\alpha(\sigma-1)}{\sigma}\right)-\left(1-\frac{\alpha(\sigma-1)}{\sigma}\right)+\frac{2}{\sigma}\right.$$
$$\left.-\left(2-t\left(\frac{\alpha(\sigma-1)}{\sigma}-1\right)\right)-t\left(\frac{\alpha(\sigma-1)}{\sigma}-1\right)\right] = 0,$$

where the first term in brackets captures changes on aggregates prices in both countries, the second (respectively the third) term is change on skilled labor revenue in the domestic (respectively foreign) country, the fourth is change on capital rent, the fifth (respectively the fifth) is changes on government revenue in the domestic (respectively foreign) country.

Changes in government revenue due to changes in the fiscal base in the two countries canceled each other out. In addition, changes in revenue in the domestic country due to an increase in the amount of tax that each remaining firm has to pay is canceled by the direct effect of the tax in both countries. The indirect effect on the aggregate level of prices is canceled out by part of changes in skilled labor revenue in the two countries. As mentioned above, all the effects cancel each other out, and the cooperative optimal tax in the origin principle is therefore a subsidy equal to :

$$t_o = -\frac{1}{\sigma}.$$

Proposition 1 *The optimal tax rate in the cooperative case is the same in the destination or origin principle; it is a subsidy equal to*

$$\hat{t} = -\frac{1}{\sigma}. \tag{41}$$

This result reflects a canonical insight from optimal taxation theory under monopolistic competition: a unit subsidy equal to the markup distortion precisely offsets the welfare loss associated with pricing above marginal cost. At $t = -\frac{1}{\sigma}$, consumer prices are aligned with marginal costs, restoring first-best efficiency in the consumption of differentiated goods. This aligns with the theoretical benchmark derived in Reinhorn (2012), and extends the classical Atkinson and Stiglitz (1976) framework to an open economy with firm-level heterogeneity and labor segmentation. Notably, the optimal tax here acts as a Pigouvian correction, not for externalities in consumption, but for pricing distortions arising from market power, and is invariant to the choice of tax principle under cooperative regimes. This neutrality under cooperation, however, breaks down under strategic tax setting, as shown in subsequent sections.

In what follows, we give the characterization of the optimal tax principle when countries

behave independently one to another. The distinction between cooperative and non-cooperative tax regimes is particularly relevant in the framework developed here because the model explicitly captures strategic interactions between countries through mobile firms, factor markets, and cross-border demand. Each government's commodity tax affects not only domestic prices but also the distribution of production and the skilled wage differential across countries. When countries act non-cooperatively, they ignore these cross-border spillovers, leading to tax policies that are strategically distorted and potentially inefficient. By contrast, the cooperative regime serves as a benchmark in which both governments internalize these externalities and jointly choose tax rates that maximize global welfare. Comparing the two regimes therefore allows us to quantify the efficiency loss from fiscal competition and to assess whether the origin or destination tax principle is more prone to welfare-reducing strategic behavior. This distinction is not merely theoretical: it directly parallels current debates on international tax coordination and harmonization, where the tension between national autonomy and global efficiency remains a central policy concern.

6 Non cooperative case

In this section, we give a characterization of the optimal tax formulae for countries acting independently. The main idea here is to observe how governments set tax levels in response to any change in tax policy abroad. More specifically, the aim is to see how close the level of the non-cooperative tax is to that of the benchmark.

6.1 Non cooperative with the destination principle

In this subsection we study the non-cooperative tax in the destination principle. The key question here is to characterize what the reaction (in terms of tax policy) will be of one government in response to a foreign policy. We bring out results in a Nash-symmetric equilibrium.

For this purpose, we examine how the domestic country h adjusts its tax level to maintain the individual level of welfare described in (8). The first derivative of the question is given by (29). This equation helps us to gain a deeper understanding on the different interactions and externalities that are needed to be taking into account by the local government when setting up its policy. Indeed, it is likely possible that, different from the cooperative case, the concern of any government in the non-cooperative case goes beyond the sole problem of resolving externalities from the monopolistic competition. Using the derivative of the results

obtained in Section (4.1), equation (29) in the destination principle becomes:

$$-\frac{L\sum_i \psi_i \alpha_i}{2\left(1+t\right)^2}\left[\left(1+t\right)\left(2-\alpha\right)+\frac{\alpha\left(\sigma-1\right)}{\sigma}+\frac{1}{\sigma}-2\right] \quad = \quad 0,$$

where the first term in the above brackets are changes on welfare due to changes on the aggregate level of prices, the second and third terms are changes on welfare due to changes of revenues (labor revenue and capital revenue respectively) and the final term been changes on welfare due to changes of the level of government lump sum transfers.

Therefore, the optimal tax response of the home country for any tax policy t^* in the foreign country, in the destination principle in a symmetric Nash equilibrium as following:

$$\bar{t}_d = -\frac{1}{\sigma}\left(\frac{1-\alpha}{2-\alpha}\right) \tag{42}$$

Proposition 2 *Under the destination principle, the best response of the domestic country to any foreign policy t^* is a subsidy equals to $\bar{t}_d = -\frac{1}{\sigma}\left(\frac{1-\alpha}{2-\alpha}\right)$. This subsidy is lower than in the cooperative case.*

Equation (42) characterizes the non-cooperative optimal commodity tax under the destination principle as $\bar{t}_d = -\frac{1}{\sigma}\left(\frac{1-\alpha}{2-\alpha}\right)$, contrasting with the cooperative benchmark tax in equation (41), $\hat{t} = -\frac{1}{\sigma}$, which fully internalizes the distortion from monopolistic competition. This deviation from the benchmark has direct implications for skilled wage determination. In the destination principle regime, as shown in equation (22), the skilled wage is inversely related to the domestic tax rate: a higher tax (or smaller subsidy) reduces product market demand, thereby diminishing firms' demand for skilled labor and, consequently, skilled wages. In equilibrium, the non-cooperative tax being a less generous subsidy than under coordinated policy, exerts downward pressure on skilled wages relative to the cooperative benchmark. The length of this wage suppression is directly related to α, the elasticity-adjusted share of skilled labor in production. As α increases, signaling a greater dependence on skilled labor, the marginal benefit of subsidization decreases thus reducing the incentive of government to offer fiscal support. This reflects a key trade-off: while subsidies can mitigate pricing distortions from monopolistic competition, they also elevate wage costs, particularly in skill-intensive sectors. This is because subsidies amplify wage costs, which in turn exacerbate price distortions under monopolistic competition. This result aligns with the broader literature that emphasizes the inefficiency of decentralized tax setting in the presence of cross-border spillovers (Mintz and Tulkens (1986); Keen et al. (1996)), but it extends those insights by introducing wage endogeneity and labor heterogeneity into the analysis. Specifically, the presence of flexible skilled wages introduces a new channel through which tax policy

affects market outcomes, and this channel is absent in frameworks assuming fixed wages or homogeneous labor (e.g., Lockwood (1993); Haufler and Pflüger (2004)). Thus, equation (42) not only quantifies the strategic deviation from the cooperative tax but also reveals how labor market composition—through α—shapes optimal policy design in open economies.

6.2 Non cooperative case with the origin principle

In the origin principle, any tax variation falls solely on goods produced in the country that set it up, hence the profit of firms that are located in that country. As we did in the previous section, we analyze the reaction of the domestic government to any tax policy in the foreign country. For this purpose we determine the tax level that set to zero equation (29).

Using the results in section (4.2) and so their derivatives, equation (29) in the origin principle writes:

$$-\frac{L\sum_i \psi_i \mu_i}{2\left(1+t\right)^2}\left[\left(1+t\right)\left(1-\alpha\right)+\left(1+\frac{\alpha\left(\sigma-1\right)}{\sigma}\right)+\frac{1}{\sigma}-\left(2-t\left(\frac{\sigma}{\alpha\left(\sigma-1\right)}-1\right)\right)\right] \quad = \quad 0,$$

where the first term in brackets is the effects of the aggregate price changes on welfare, the two following terms are the effects of changes of revenue on welfare and the last term is the effects of changes of government lump sum transfers on welfare.

Changes in revenue due to the increase in the amount collected from firms are offset by part of the change on the aggregate price level. Part of changes in revenue due to firms relocation is also canceled out by part of the changes in the aggregate price level. The remaining effects are those that form the structure of the non-cooperative tax.

Therefore, the best response to a foreign policy in a symmetric Nash equilibrium is the following tax level:

$$\bar{t}_o = -\frac{1}{\sigma}\frac{1-\alpha}{\frac{\sigma}{\alpha(\sigma-1)}-\alpha} \tag{43}$$

Proposition 3 *According to the origin principle, the best response of the domestic country h to any foreign policy t^* is a subsidy equal to $\bar{t}_o = -\frac{1}{\sigma}\frac{1-\alpha}{\frac{\sigma}{\alpha(\sigma-1)}-\alpha}$. This subsidy is lower than in the cooperative case.*

Equation (43) characterizes the optimal non-cooperative tax policy under the origin principle, revealing that strategic tax-setting in decentralized regimes leads to a subsidy that is systematically lower than the cooperative benchmark defined in equation (41). This divergence arises due to the endogenous response of firm location to tax differentials, a

feature unique to the origin principle. As taxes apply to production, higher domestic taxation induces firm relocation, which in turn depresses domestic demand for skilled labor and exerts downward pressure on skilled wages. Consequently, the government internalizes not only the domestic distortions from monopolistic competition but also the cross-border wage spillovers generated by mobile firms. In contrast to the cooperative setting—where subsidies fully correct for monopolistic pricing—non-cooperative governments under the origin principle offer more limited support, resulting in lower skilled wages relative to the first-best. This wage asymmetry is amplified when skilled labor accounts for a smaller share of production income, highlighting the central role of labor composition in shaping fiscal efficiency. The result underscores that wage sensitivity to tax policy is greater under the origin principle, and optimal unilateral policy tends to be more distortionary in environments characterized by mobile firms and flexible skilled wages.

This result contributes to and refines a growing body of literature on tax competition in integrated markets. Classical neutrality results—such as those by Grossman (1980) and Lockwood (1993)—demonstrated that, under idealized conditions (e.g., fixed wages, homogeneous labor, and symmetric countries), the origin and destination principles yield equivalent efficiency outcomes. However, equation (43) challenges this neutrality by introducing wage endogeneity and firm mobility into the analysis. In doing so, it aligns more closely with the insights of Mintz and Tulkens (1986) and Keen et al. (1996), who emphasized the inefficiencies induced by decentralized tax setting under the origin principle. Our result extends their framework by showing that wage flexibility, particularly for skilled labor, significantly amplifies these inefficiencies due to tax-induced relocation effects.

Proposition 4 *When the share of firm revenue allocated to the use of skilled labor is less than half, that is,*

$$\frac{\alpha\left(\sigma-1\right)}{\sigma}<\frac{1}{2}, \tag{44}$$

, then at equilibrium, taxes in the non-cooperative destination principle are smaller than those in the non-cooperative origin principle.

Proposition 4 establishes a critical threshold condition that determines the relative efficiency of origin-based versus destination-based taxation in non-cooperative settings. Specifically, it states that when the share of production income accruing to skilled labor—captured by the expression $\frac{\alpha(\sigma-1)}{\sigma}$ —falls below one-half, the optimal unilateral subsidy under the origin principle is smaller than that under the destination principle. This implies that governments in economies with relatively low skilled-labor intensity have stronger incentives to subsidize consumption rather than production, making destination-based taxation more effective in mitigating monopolistic distortions without triggering inefficient firm relocation.

This result carries significant theoretical weight because it identifies the composition of labor income as a sufficient statistic for ranking tax regimes. When skilled labor constitutes a minor share of production costs, the marginal welfare gain from preserving skilled labor income through origin-based tax relief is limited. In such settings, any tax-induced firm relocation imposes a disproportionate welfare cost relative to the potential gains from wage flexibility. Conversely, under the destination principle—where firm location is invariant to tax changes—governments can adjust consumption taxes without inducing relocation, making destination-based subsidies more robust in protecting domestic welfare. This insight challenges the neutrality results of earlier models, such as Lockwood (1993), by showing that labor heterogeneity and wage endogeneity break the equivalence between tax principles in a meaningful and policy-relevant way.

Proposition 4 also serves as a bridge between conflicting strands of the literature. While Haufler and Pflüger (2004) argue that destination-based taxation generally Pareto-dominates origin-based taxation under monopolistic competition, our results nuance this conclusion. We show that the superiority of either principle depends not only on market structure but also on the interaction between tax incidence, firm mobility, and labor market segmentation. In doing so, Proposition 4 advances the theory of international tax design by embedding it within a richer model of production that captures the asymmetric burden of taxation across heterogeneous labor groups.

Our framework also sheds light on an increasingly relevant concern in international tax policy—the phenomenon of "skill downgrade." This refers to the possibility that commodity tax reforms, particularly those that raise producer prices or trigger firm relocation, may disproportionately harm skilled workers by reducing the demand for high-skilled labor and thus lowering their wages. Because our model explicitly incorporates labor market segmentation and endogenously determined skilled wages, it captures this mechanism rigorously. We show that an increase in commodity taxation —especially under the origin principle—leads to a sharper decline in skilled wages, while unskilled workers remain unaffected due to wage rigidity. This asymmetric incidence highlights the distributional risks of poorly calibrated tax policies, particularly in low skill-intensive sectors or countries. The threshold condition we derive based on the share of production income accruing to skilled labor thus serves not only as a guide for welfare-maximizing tax regimes, but also as an indicator of where the risk of skill downgrade is most acute. In this sense, our results contribute to both the efficiency and equity dimensions of international tax coordination.

7 Empirical Relevance and a Quantitative Illustration

We now calibrate the model using EU KLEMS Labour Accounts data (Stehrer et al., 2019) to assess where actual economies stand relative to the 50% threshold we previously identified in Proposition 4. The calibration proceeds in two steps: first at the aggregate (whole economy) level, then at the sectoral level. This exercise serves three purposes. First, it tests the empirical relevance of our theoretical threshold. Second, it reveals substantial cross-country and cross-sector heterogeneity in skill intensity, with direct implications for the optimal tax principle. Third, it demonstrates that our results are not knife-edge but vary continuously with the underlying labor market structure.

7.1 Wage flexibility and wage rigidity test

This subsection describes the data sources, variable construction, and econometric methodology used to test the key behavioral assumption underlying our theoretical model: that skilled workers exhibit flexible wages while unskilled workers face wage rigidity. Establishing this heterogeneity empirically is crucial for the validity of our theoretical predictions regarding the differential welfare effects of origin- versus destination-based taxation.

7.1.1 Data Sources and Sample Construction

We draw on the EU KLEMS 2019 release, a comprehensive database providing industry-level measures of output, inputs, and productivity for European Union member states and other major economies (Stehrer et al., 2019). The database's distinctive advantage lies in its detailed cross-classification of labor inputs by educational attainment, enabling us to construct skill-specific wage and employment measures that align with our theoretical distinction between skilled (high-educated) and unskilled (low-educated) workers.

Our analysis focuses on eleven countries and each country, we extract two core variables from the EU KLEMS labor files:

- Employment shares (H_{shares}) : The share of total hours worked accounted for by each labor type, defined by the intersection of gender (male/female), age group (15-29, 30-49, 50+), and educational attainment (low, medium, high). Education levels follow the International Standard Classification of Education (ISCED): low (ISCED 0-2), medium (ISCED 3-4), and high (ISCED 5-8).

- Labor compensation shares (W_{shares}) : The corresponding share of total labor compensation accruing to each labor type.

Following what is done in the literature(e.g., Katz and Autor, 1999a or Acemoglu and Autor, 2011), we aggregate across gender and age groups within each education level to obtain annual time series of the employment share s_{it}^E and wage share s_{it}^W for education group $i \in \text{Low}, \text{Medium}, \text{High}$ in year t. Our sample spans the period $2008 - 2017$, providing a balanced panel of ten annual observations per country, sufficient to capture cyclical variation while maintaining data consistency across all eleven economies.

The relative wage for each education group is then constructed as:

$$w_{it} = \frac{s_{it}^W}{s_{it}^E}$$

This ratio measures the average wage of workers in education group i relative to the economy-wide average wage. Values exceeding unity indicate above-average compensation, reflecting a skill premium, while values below unity indicate below-average compensation.

7.1.2 Econometric Framework

To test for differential wage flexibility across skill groups, we estimate the elasticity of relative wages with respect to employment shares. The intuition is straightforward: if wages are flexible, an increase in the relative supply of a particular skill group should exert downward pressure on its relative wage, implying a negative elasticity. Conversely, if wages are rigid, this relationship should be attenuated or absent.

Our baseline specification takes the form:

$$\ln w_{ict} = \beta_i \ln E_{ict} + \alpha_c + \gamma_t + \varepsilon_{ict} \tag{45}$$

where w_{ict} is the relative wage of education group i in country c and year t, E_{ict} is the corresponding employment share, α_c are country fixed effects capturing time-invariant institutional and structural differences across labor markets, and γ_t are year fixed effects absorbing common macroeconomic shocks (e.g., the Eurozone debt crisis, the post-2008 recovery). The parameter of interest, β_i, represents the elasticity of relative wages with respect to employment share for group i. We estimate equation (45) separately for each education group using panel ordinary least squares with two-way fixed effects.

7.1.3 Testing for Cross-Group Differences

While separate regressions by education group provide intuitive estimates of group-specific elasticities, they do not directly test whether these elasticities differ statistically across groups. We therefore complement our analysis with a pooled regression interacting employment shares with education group dummies:

$$\ln w_{ict} = \beta_0 \ln E_{ict} + \beta_1(\ln E_{ict} \times \text{High}_{ict}) + \beta_2(\ln E_{ict} \times \text{Medium}_{ict}) + \alpha_c + \gamma_t + \delta_i + \varepsilon_{ict}, \quad (46)$$

where High_{ict} and Medium_{ict} are indicators for high and medium education, respectively (low education serves as the omitted category). The coefficient β_0 captures the wage elasticity for low-educated workers, while β_1 and β_2 measure the differential elasticity for high- and medium-educated workers relative to the low-educated baseline. A negative and significant β_1 would indicate that high-skilled wages respond more elastically to supply shifts, consistent with greater flexibility. Equation 46 includes education group fixed effects δ_i to absorb level differences in relative wages, ensuring that the interactions capture slope heterogeneity rather than level effects.

7.1.4 Descriptive

Table 1: Relative wages by education

	Low	Medium	High
Relative wage	0.651	0.934	1.381
	(0.177)	(0.075)	(0.252)
Observations	110	110	110

Standard deviations in parentheses.

Table 1 reports average relative wages and labor shares by education group. A clear skill premium emerges: high-educated workers earn on average 38% more than the reference wage, while low-educated workers earn about 35% less. Medium-educated workers lie in between. This ranking is consistent with standard evidence on wage heterogeneity across skill groups.

The distribution of labor shares also differs markedly across education levels, indicating that the composition of employment varies systematically with skills. These patterns suggest that labor-market outcomes are strongly segmented by education.

7.1.5 Main Results

Table 2: The Heterogeneous Relationship Between Employment and Wages by Education Level

Dependent Variable:	(1)	(2)	(3)	(4)
Log Real Wage	**High**	**Medium**	**Low**	**Pooled**
Log Employment	−0.0308	0.7165**	0.0722	−0.0556
	(0.3709)	(0.3052)	(0.2579)	(0.043)
	[0.934]	[0.021]	[0.780]	[0.201]
Log Employment × High Education				0.2345***
				(0.014)
				[0.000]
Log Employment × Medium Education				0.1164***
				(0.015)
				[0.000]
Constant				−0.3336***
				(0.128)
				[0.009]
Country Fixed Effects	*Yes*	*Yes*	*Yes*	*Yes*
Year Fixed Effects	*Yes*	*Yes*	*Yes*	*Yes*
Observations	110	110	110	330
R-squared	0.001	0.246	0.004	0.677
F-test for Poolability				
F-statistic	50.34***	< 48.42***	167.97***	
P-value	[0.000]	[0.000]	[0.000]	
Implied Total Effects				
High Education				0.1789***
Medium Education				0.0608
Low Education				−0.0556

Notes: Heteroskedasticity-robust standard errors in parentheses (HC3 in column 4; clustered at country level in columns 1-3). P-values in brackets. All specifications include country and year fixed effects. Columns 1-3 estimated using PanelOLS with entity and time effects. Column 4 estimated using OLS with country and year indicators. The omitted education category in column 4 is Low Education. Implied total effects calculated from column 4 coefficients. Statistical significance for implied effects based on linear combination tests. *** p<0.01, ** p<0.05, * p<0.10.

Table 2 reports two-way fixed-effects estimates of the elasticity of wages to employment by education group. Wages of low-education workers display no statistically significant response to employment fluctuations, consistent with institutional wage rigidity. In contrast, wages of more educated workers are significantly more responsive to labor-demand conditions, with the largest elasticity for the high-education group in the pooled interaction specification. Quantitatively, a 10% increase in employment is associated with a wage increase of about 1.8% for highly educated workers and around 0.6% for medium educated workers, while the

corresponding response for low-education workers is close to zero. This pattern provides direct evidence of labor-market segmentation and supports the key mechanism of the model, in which adjustments to firm relocation operate primarily through the flexible skilled wage, high and medium educated workers, while the unskilled wage remains rigid.

7.2 Defining Skilled Labor in the Model

A critical measurement issue in calibrating our model is the empirical demarcation between skilled and unskilled labor. Our production function $F(\ell_s, \ell_u) = A\ell_s^{\alpha}\ell_u^{1-\alpha}$ assumes that skilled labor (ℓ_s) is characterized by flexible wages, complementarity with capital, and adaptability to firm-specific productivity shocks. A substantial body of labor economics literature establishes that these characteristics emerge not at the tertiary threshold, but at the completion of secondary education, which confers foundational numeracy, literacy, and problem-solving skills (Autor, 2014; Goldin & Katz, 2008; OECD, 2019). Workers with less than upper-secondary education are predominantly employed in routine manual occupations, face binding wage floors, and exhibit low substitutability with capital–precisely the properties we assign to unskilled labor.

We therefore adopt the **broad definition (ISCED 3-8)** as our preferred skill classification, grouping workers with at least upper-secondary education. This aggregation is consistent with standard practice in macroeconomic and growth-accounting analyses that use education as a proxy for skill, and captures the economically relevant margin of human capital while preserving cross-country comparability. Results using the narrow definition (tertiary only, ISCED 5-8) are reported in Appendix A for transparency and comparability with earlier studies. However, in our view this narrow definition understates the effective skill intensity of modern production and would misclassify as "unskilled" large segments of the workforce that possess the flexibility our model associates with skilled labor.

7.3 The 50% Threshold: A Structural Property

Before turning to the data, it is essential to establish that the 50% threshold is a structural feature of the model, not an artifact of parameter selection. Figure 1 plots the welfare difference between non-cooperative origin and destination equilibria, $\Delta W = W_o - W_d$, as a function of the skilled labor share α, holding all other parameters at their symmetric baseline values ($\sigma = 4$, $L = L^*$, $\psi_s = \psi_s^*$, $\mu_i = 1$, $\bar{K} = 1$).[9]

[9] The elasticity of substitution $\sigma = 4$ is within the range commonly used in the trade and macro literature (e.g., Anderson and van Wincoop, 2004; Broda and Weinstein, 2006).

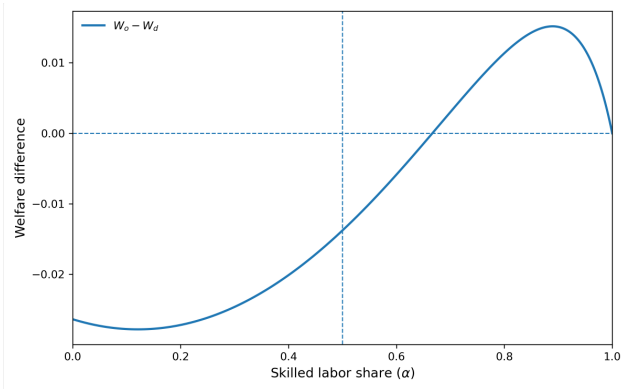

Figure 1: Welfare Comparison: Origin vs. Destination

Note: Welfare difference computed under symmetric non-cooperative equilibrium. Parameters are $\sigma = 4$, $\mu = 1$, and $K = 1$.

Figure 1 yields three immediate insights. First, the welfare ranking of tax principles is *monotonic* in α: as the economy becomes more skill-intensive, the relative performance of the origin principle improves continuously. Second, the crossing point occurs precisely at $\alpha = 0.5$, confirming Proposition 4. Third, the function is smooth with no discontinuities; there is no reversal or instability at other parameter values. This monotonicity implies that measurement error in α affects the *magnitude* of the welfare gain from switching regimes, but not the *sign* for countries sufficiently far from the threshold. The 50% rule is therefore robust and operational.[10]

7.4 Aggregate Results

In this section we use equation (13) to calibrate α for each country as following:

$$\alpha = \frac{\omega_k \ell_{s,k}}{\omega_k \ell_{s,k} + \ell_{u,k}},$$

and with this value of α, we compute optimal taxes given by equations (42) and (43).

Following the results in the previous subsection, We define skilled labor as the population with at least upper-secondary education, grouping individuals with medium and high education levels (ISCED 3-8). This definition reflects a threshold view of human capital, according to which productivity-relevant skills emerge once basic secondary education is completed. Workers with upper-secondary or tertiary education possess the cognitive, technical, and task-specific competencies required to effectively operate modern production

[10]Precisely, the ranking depends on the elasticity of substitution σ which determines the threshold $\alpha^\star = \frac{\sigma}{2(\sigma-1)}$: as σ increases this threshold approaches 0.5.

technologies and to adapt to capital- and knowledge-intensive processes. Table 3 reports α under our preferred broad definition for 11 EU member states, along with the corresponding optimal non-cooperative tax rates $\bar{t}_o$ and $\bar{t}_d$ (with $\sigma = 4$) and the threshold status.

Table 3: Skilled labor share (α) under broad definition (ISCED 3-8), EU countries, 2019

Country	α	$\bar{t}_o$	$\bar{t}_d$	Threshold	Implied preferred regime
Germany	0.941	-0.031	-0.0138	0.706	Origin
Denmark	0.875	-0.048	-0.0278	0.656	Origin
France	0.858	-0.051	-0.0310	0.644	Origin
Hungary	0.941	-0.031	-0.0138	0.706	Origin
Poland	0.971	-0.018	-0.0071	0.728	Origin
Romania	0.938	-0.032	-0.0145	0.704	Origin
Austria	0.930	-0.035	-0.0163	0.698	Origin
Sweden	0.909	-0.041	-0.0209	0.681	Origin
Italy	0.749	-0.061	-0.0501	0.562	Origin
Spain	0.760	-0.060	-0.0484	0.570	Origin
Netherlands	0.797	-0.058	-0.0422	0.598	Origin

Notes: Skilled labor defined as workers with at least upper-secondary education (ISCED 3-8). α computed as share of labor compensation accruing to skilled workers. Tax rates $\bar{t}_o$ and $\bar{t}_d$ are optimal non-cooperative subsidies under origin and destination principles, $\sigma = 4$. All countries lie above the 50% threshold, implying welfare superiority of the origin principle. Source: EU KLEMS Labour Accounts (Stehrer et al., 2019).

Table 3 reports the cost of skilled labor as a ratio of the total labor cost such as to test our threshold condition in Proposition 4. Empirically, wage differentials and productivity outcomes tend to exhibit a sharper discontinuity between low-educated workers and those with at least secondary education than between medium- and high-educated workers, supporting a binary distinction between skilled and unskilled labor.

The results are striking. Under the broad definition, *every* EU member state in our sample exhibits $\alpha > 0.5$. The average α is 0.87, and the lowest value (Italy, 0.749) remains substantially above the theoretical threshold. This implies that, conditional on our model, the origin principle would yield higher welfare than the current destination-based VAT system in *all* countries analyzed. Moreover, the welfare gain is not marginal: at $\alpha = 0.75$, Figure 1 shows a positive and economically significant gap between origin and destination welfare.

This finding challenges the conventional wisdom that destination-based taxation is inherently superior in integrated markets (Haufler and Pflüger, 2004). It also provides a structural explanation for the long-standing reluctance of some EU member states to harmonize VAT rates around a pure destination system: countries with high skill intensity *benefit* from origin-based features that allow them to attract firms and leverage skilled wage flexibility.

8 Conclusion

This paper contributes to the long-standing debate on the appropriate principle for indirect taxation in open economies, destination versus origin, by embedding the analysis within a general equilibrium model that features monopolistic competition, international firm mobility, and heterogeneous labor markets. In contrast to the standard framework that assumes homogeneous agents and uniform tax pass-through, we highlight how labor market segmentation, especially wage rigidity for unskilled workers and flexible wages for skilled workers, can fundamentally alter the incidence and efficiency of tax policies.

Our central finding is that neither the origin nor the destination principle unconditionally dominates in terms of welfare outcomes. Instead, their relative desirability hinges on the share of production income accruing to skilled labor. Specifically, when this share exceeds 50%, the origin principle Pareto dominates; otherwise, the destination principle yields superior outcomes. This threshold result offers a tractable and empirically grounded decision rule for policymakers, anchored in observable labor market characteristics. It provides a concrete criterion for identifying the optimal tax regime across sectors or jurisdictions with varying labor compositions. These findings qualify conventional wisdom that destination-based taxation is inherently superior in a globalized context. Although the destination principle may support consumption neutrality and minimize trade distortions, it can be welfare inferior when tax-induced firm relocation interacts with wage adjustments in the skilled labor market. The origin principle, often viewed as trade-distorting, may in fact improve welfare when firms rely heavily on mobile, high-wage labor and when fiscal spillovers impact foreign wage levels asymmetrically.

Our framework also revisits earlier conclusions from Lockwood (1993), Mintz and Tulkens (1986), and Keen et al. (1996), and shows how their policy implications change when wage endogeneity and labor segmentation are introduced. By doing so, we contribute to the emerging literature that seeks to reconcile theoretical tax neutrality results with real-world frictions, such as labor market institutions and cross-border firm dynamics. From a policy perspective, this paper offers practical guidance. The share of skilled labor, often measurable through input-output tables or wage bill statistics, can serve as a sufficient statistic to identify when origin- or destination-based taxation improves welfare. This has implications for international tax coordination, especially in regional economic blocs where sectoral specialization and labor heterogeneity vary widely.

There are several avenues for future research. Our analysis abstracts from capital taxation, administrative costs of enforcement, transport costs and sectoral heterogeneity in markups and tax sensitivity. These factors could interact meaningfully with labor market

segmentation to shape tax incidence. Finally, while our analysis primarily emphasizes the efficiency and neutrality properties of optimal commodity taxation under monopolistic competition and international firm mobility, it is important to acknowledge the distributional implications of the resulting fiscal structure. The optimal policy entails a subsidy on the consumption of differentiated goods, with revenues redistributed via uniform lump-sum transfers. In this framework, higher-income individuals, who allocate a larger share of expenditure to such goods, derive greater absolute benefit from the subsidy. This raises potential equity concerns, highlighting a trade-off between efficiency gains and the regressivity of consumption-based redistribution. While distributional objectives are beyond the scope of this paper, incorporating agent heterogeneity or equity-based welfare criteria could enrich the analysis and offer insights into the broader political and normative feasibility of international tax coordination.

A Index: Concavity conditions

A.1 Destination principle

Let define by

$$f(t) = -t + \frac{\alpha\left(1+t\right)}{1+\frac{1+t^*}{1+t}\frac{L^*\sum_i \psi_i^*\mu_i}{L\sum_i \psi_i\mu_i}} - \frac{\sigma-1}{\sigma}\frac{\alpha}{1+\frac{L^*\psi_i^*}{L\psi_s}} - \frac{1}{2\sigma},$$

the function that draw the sign of the utility in the destination non cooperative case. We will study the variation of this function.

For every given t, and $h \leq 0$

$$g(h) = f(t+h) - f(t) = -h + \frac{\alpha\gamma + \alpha h}{1+\frac{\theta}{\gamma+h}} - \frac{\alpha\gamma}{1+\frac{\theta}{\gamma}},$$

where $\gamma = 1+t$ and $\theta = \left(1+t^*\right)\frac{L^*\sum_i \psi_i^*\alpha_i}{L\sum_i \psi_i\alpha_i}$

$$\begin{aligned}
\lim_{h\to+\infty} g(h) &= \lim_{h\to+\infty} h\left[-1+\frac{\alpha\left(1+\frac{\gamma}{h}\right)}{1+\frac{\theta}{\gamma+h}}\right] - \frac{\alpha\gamma}{1+\frac{\theta}{\gamma}} = -\infty \\
\lim_{h\to-\infty} g(h) &= \lim_{h\to-\infty} h\left[-1+\frac{\alpha\left(1+\frac{\gamma}{h}\right)}{1+\frac{\theta}{\gamma+h}}\right] - \frac{\alpha\gamma}{1+\frac{\theta}{\gamma}} = +\infty \\
\lim_{h\to 0} g(h) &= \lim_{h\to 0} \frac{\alpha\gamma}{1+\frac{\theta}{\gamma}} - \frac{\alpha\gamma}{1+\frac{\theta}{\gamma}} = 0
\end{aligned}$$

Moreover, for any given t,

$$g'(h) = -1 + \alpha\frac{1+\frac{2\theta}{\gamma+h}}{\left(1+\frac{\theta}{\gamma+h}\right)^2}$$

$$\begin{aligned}
g'(h) = 0 \Longleftrightarrow \left(1+\frac{\theta}{\gamma+h}\right)^2 - \alpha\left(1+\frac{2\theta}{\gamma+h}\right) &= 0 \\
\Longleftrightarrow X^2 - 2X\left(1-\alpha\right) + \left(1-\alpha\right) &= 0,
\end{aligned}$$

where $X = \frac{\theta}{\gamma+h}$.
The discriminant of the above equation is given by:

$$\begin{aligned}
\Delta &= 4\left(1-\alpha\right)^2 - 4\left(1-\alpha\right) \\
&= -4\alpha\left(1-\alpha\right) < 0
\end{aligned}$$

$g'(h) < 0$, then g is decreasing, meaning f is strictly decreasing in $]-\infty;+\infty[$, specially in $]-1;1[$.

A.2 Origin principle

Let define by

$$f(t) = \frac{\sigma}{\sigma-1}(1+t)(1+C)^2 - \frac{\sigma}{\alpha(\sigma-1)}\frac{(1+t)(1+C)^2}{(1-\sigma)(1+B)}$$
$$+\alpha(1+t)(1+C)\left(1+\frac{\sigma}{\alpha(\sigma-1)}C\right)+t\left(1-\frac{L^*\sum_i\psi_i^*\mu_i}{L\sum_i\psi_i\mu_i}\right)\left(1-\frac{\sigma}{\sigma-1}C\right)$$
$$\left(1-\frac{L^*\sum_i\psi_i^*\mu_i}{L\sum_i\psi_i\mu_i}\right)\left(1-\frac{\alpha(\sigma-1)}{\sigma}+\frac{(1+B)\left[\frac{\sigma}{\alpha(\sigma-1)}\frac{B-C}{1+B}-1\right]}{2\sigma}\right)$$

the function that draws the sign of the utility in the origin non-cooperative case. We will study the variation of this function.

$$\begin{aligned}\lim_{t\to-1} f(t) &= \lim_{t\to-1}(1+t)\left[\frac{\sigma}{\sigma-1}\frac{\alpha(\sigma-1)+1}{\alpha(\sigma-1)}+\alpha+\left(1-\frac{L^*\sum_i\psi_i^*\mu_i}{L\sum_i\psi_i\mu_i}\right)\right]\\ &\quad-\left(1-\frac{L^*\sum_i\psi_i^*\mu_i}{L\sum_i\psi_i\mu_i}\right)\left(\frac{\alpha(\sigma-1)}{\sigma}+\frac{1}{2\sigma}\right)\\ &= +\infty\end{aligned}$$

In addition, $f(0) < +\infty$, so there is an interval in $]-1;1[$ where f is strictly decreasing.

B Firm's marginal cost expression

Proof. To produce one unit of output, a given firm chooses $\ell_{s,k},\ \ell_{u,k}$ to solve the following problem:

$$\begin{cases}\min_{\ell_{s,k},\ell_{u,k}} C = \omega_k\ell_{s,k}+\ell_{u,k}\\ \text{subject to}: A\ell_{s,k}^{\alpha}\ell_{u,k}^{1-\alpha}=1.\end{cases}$$

FOC leads to $\ell_{u,k} = \frac{1-\alpha}{\alpha}\omega_k\ell_{s,k}$ Substituting this equation back into the production constraint $A\ell_{s,k}^{\alpha}\ell_{u,k}^{1-\alpha}=1$ leads to

$$\ell_{s,k} = A^{-1}\left(\frac{1-\alpha}{\alpha}\omega_k\right)^{-(1-\alpha)} \tag{47}$$

Therefore the marginal cost that is the minimal cost of producing one additional output is given by: $C(1) = \omega_k \ell_{s,k} + \ell_{u,k} = \frac{\omega_k \ell_{s,k}}{\alpha}$.

Substituting $\ell_{s,k}$ from 47 into the marginal cost expression leads to:

$$C(1) = \frac{\omega_k^{\alpha}}{A\,(\alpha)^{\alpha}\,(1-\alpha)^{1-\alpha}}.$$

■

C Robustness checks

Table 4: Baseline Panel Regressions by Education Level

	High Education	Medium Education	Low Education
$\ln(h_{share})$	-0.9981***	-0.9963***	-0.9888***
	(0.0006)	(0.0014)	(0.0049)
Constant	1.6342***	1.8437***	0.5227***
	(0.0009)	(0.0028)	(0.0052)
Observations	3960	3960	3960
Countries	11	11	11
Years	2008-2017	2008-2017	2008-2017
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
R^2 (within)	0.3917	0.3975	0.3902

Clustered standard errors at the country level in parentheses.
*** $p < 0.01$.

Table 5: Robustness check: Aggregate Skilled Labor Share α and Optimal Taxes $\bar{t}_o$ and $\bar{t}_d$ ($\sigma = 2.5$)

Country	α	$\bar{t}_o$	$\bar{t}_d$	Threshold	Preferred regime
Germany	0.941	-0.028	-0.0222	0.565	Origin
Denmark	0.875	-0.049	-0.0445	0.525	Origin
France	0.858	-0.052	-0.0496	0.515	Origin
Hungary	0.941	-0.028	-0.0221	0.565	Origin
Poland	0.971	-0.016	-0.0114	0.582	Origin
Romania	0.938	-0.029	-0.0232	0.563	Origin
Austria	0.930	-0.032	-0.0261	0.558	Origin
Sweden	0.909	-0.039	-0.0335	0.545	Origin
Italy	0.749	-0.068	-0.0802	0.450	Destination
Spain	0.760	-0.067	-0.0775	0.456	Destination
Netherlands	0.797	-0.063	-0.0675	0.478	Destination

References

Acemoglu, D. and Autor, D. (2011). Chapter 12 - skills, tasks and technologies: Implications for employment and earnings. volume 4 of *Handbook of Labor Economics*, pages 1043–1171. Elsevier.

Aiura, H. and Ogawa, H. (2023). Does e-commerce ease or intensify tax competition? destination principle versus origin principle. *International Tax and Public Finance*, 31(3):702–735.

Antoniou, F., Hatzipanayotou, P., and Tsakiris, N. (2019). Destination-based vs. origin-based commodity taxation in large open economies with unemployment. *Economica*, 86(341):67–86.

Asplund, M., Friberg, R., and Wilander, F. (2007). Demand and distance: Evidence on cross-border shopping. *Journal of Public Economics*, 91(1):141–157.

Atkinson, A. B. and Stiglitz, J. E. (1976). The design of tax structure: Direct versus indirect taxation. *Journal of Public Economics*, 6(1-2):55–75.

Autor, D. H. (2014). Skills, education, and the rise of earnings inequality among the "other 99 percent". *Science*, 344(6186):843–851.

Behrens, K., Hamilton, J. H., Ottaviano, G. I., and Thisse, J.-F. (2009). Commodity tax competition and industry location under the destination and the origin principle. *Regional Science and Urban Economics*, 39(4):422–433.

Burstein, A., Lein, S., and Vogel, J. (2024). Cross-border shopping: Evidence and welfare implications for switzerland. *Journal of International Economics*, 152(C):None.

Costinot, A. and Rodríguez-Clare, A. (2014). Trade theory with numbers: Quantifying the consequences of globalization. In Gopinath, G., Helpman, E., and Rogoff, K., editors, *Handbook of International Economics*, volume 4, pages 197–261. Elsevier.

Dixit, A. K. and Stiglitz, J. E. (1977). Monopolistic Competition and Optimum Product Diversity. *American Economic Review*, 67(3):297–308.

Goolsbee, A. (2001). The implications of electronic commerce for fiscal policy (and vice versa). *Journal of Economic Perspectives*, 15(1):13–23.

Grossman, G. M. (1980). Border tax adjustments: Do they distort trade? *Journal of International Economics*, 10(1):117–128.

Haufler, A. and Pflüger, M. (2004). International Commodity Taxation under Monopolistic Competition. *Journal of Public Economic Theory*, 6(3):445–470.

Helpman, E. and Krugman, P. R. (1985). *Market Structure and Foreign Trade: Increasing Returns, Imperfect Competition, and the International Economy.* MIT Press, Cambridge, MA.

Katz, L. and Autor, D. (1999a). Changes in the wage structure and earnings inequality. In Ashenfelter, O. and Card, D., editors, *Handbook of Labor Economics*, volume 3, Part A, chapter 26, pages 1463–1555. Elsevier, 1 edition.

Katz, L. F. and Autor, D. H. (1999b). Chapter 26 - changes in the wage structure and earnings inequality. volume 3 of *Handbook of Labor Economics*, pages 1463–1555. Elsevier.

Keen, M. and Lahiri, S. (1998). The comparison between destination and origin principles under imperfect competition. *Journal of International Economics*, 45(2):323–350.

Keen, M., Smith, S., Baldwin, R. E., and Christiansen, V. (1996). The future of value added tax in the european union. *Economic Policy*, 11(23):375–420.

Krugman, P. (1980). Scale economies, product differentiation, and the pattern of trade. *American Economic Review*, 70(5):950–59.

Krugman, P. R. (1979). Increasing returns, monopolistic competition, and international trade. *Journal of International Economics*, vol. 9(4)(3):469–479.

Lockwood, B. (1993). Commodity tax competition under destination and origin principles. *Journal of Public Economics*, 52:141–162.

Lockwood, B. (2001). Tax competition and tax co-ordination under destination and origin principles: a synthesis. *Journal of Public Economics*, 81(2):279–319.

McCracken, S. (2015). The choice of commodity tax base in the presence of horizontal foreign direct investment. *International Tax and Public Finance*, 22(5):811–833.

Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica*, 71(6):1695–1725.

Mintz, J. and Tulkens, H. (1986). Commodity tax competition between member states of a federation: equilibrium and efficiency. *Journal of Public Economics*, 29(2):133–172.

Moriconi, S. and Sato, Y. (2009). International commodity taxation in the presence of unemployment. *Journal of Public Economics*, 93(7-8):939–949.

Reinhorn, L. (2012). Optimal taxation with monopolistic competition. *International Tax and Public Finance*, 19(2):216–236.

Statista (2024a). E-commerce as percentage of total retail sales worldwide from 2021 to 2027.

Statista (2024b). Retail e-commerce sales worldwide from 2014 to 2027.

Stehrer, R. A. B., Jäger, K., Reiter, O., and Schwarzhappel, M. (2019). Industry level growth and productivity data with special focus on intangible assets. *wiiw Statistical Report No. 8.*